

Problem Set 3

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Exercise 1

(a) The Lagrange Basis Polynomials of degree n are defined as:

$$L_{n,j}(x) = \prod_{i=0, i \neq j}^n \frac{x - x_i}{x_j - x_i}$$

and we know the Interpolating Polynomial

$$P_n(x) = \sum_{i=0}^n f(x_i) L_{n,i}(x)$$

For the given three data points: $\{(x_0, y_0), (x_1, y_1), (x_2, y_2)\} = \{(-1, 1/2), (0, 1), (1, 2)\}$

We calculate the Lagrange Polynomials:

$$\begin{aligned} L_{2,0}(x) &= \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-0)(x-1)}{(-1-0)(-1-1)} = \frac{x(x-1)}{2} \\ L_{2,1}(x) &= \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x+1)(x-1)}{(0+1)(0-1)} = 1 - x^2 \\ L_{2,2}(x) &= \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x+1)(x-0)}{(1+1)(1-0)} = \frac{x(x+1)}{2} \end{aligned}$$

So the Lagrange form of the Polynomial $P_2(x)$ will be:

$$\begin{aligned} P_2(x) &= \frac{1}{2} \left(\frac{x^2 - x}{2} \right) + 1(1 - x^2) + 2 \left(\frac{x^2 + x}{2} \right) \implies \\ P_2(x) &= \frac{x^2 - x}{4} + 1 - x^2 + x^2 + x \end{aligned}$$

(b) Simplified further: $P_2(x) = \frac{1}{4}x^2 + \frac{3}{4}x + 1$

For $f(x) = 2^x$, we can pick $x = 1/2$ since $2^{1/2} = \sqrt{2}$ which is the value we want to evaluate.

$$P_2(1/2) = \frac{1}{4} \left(\frac{1}{2} \right)^2 + \frac{3}{4} \left(\frac{1}{2} \right) + 1 = \frac{1}{16} + \frac{3}{8} + 1 = \frac{23}{16} \approx 1.4375$$

(c) Interpolation Error: For $x \in [a, b]$, there exists a point $\xi \in [a, b]$ such that:

$$f(x) = P_n(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1)(x - x_2)$$

where $\frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$ is the error term.

So the absolute value of this difference $|f(x) - P_n(x)|$ will be the error and so calculating the terms $(x - x_0)(x - x_1)(x - x_2)$ we get $= |-\frac{3}{8}| = \frac{3}{8}$

Calculating the derivatives:

$$f'(x) = 2^x \ln 2$$

$$f''(x) = 2^x (\ln 2)^2$$

$$f'''(x) = 2^x (\ln 2)^3$$

The maximum error on $[-1, 1]$ of $|f^{(3)}(\xi)|$ occurs when $\xi = 1$: $|f^{(3)}(\xi)| \leq 2^1 (\ln 2)^3$
 Dividing this term by $3!$: $\frac{2(\ln 2)^3}{6} = \frac{(\ln 2)^3}{3}$

Therefore, the interpolation error is

$$|f(x) - P_2(x)| \leq \frac{2(\ln 2)^3}{6} \cdot \frac{3}{8} = \frac{(\ln 2)^3}{8} \approx 0.0416$$

Exercise 2

(a) In order to define a piecewise quadratic interpolant of $n+1$ data points, we need n subsets and since we need three coefficients to define this we will need $3n$ coefficients. Conditions on s_i and derivatives should be implied:

- The first derivative has to be continuous at $n-1$: $s'_{i-1}(x) = s'_i(x)$, to ensure smoothness
- Each piecewise polynomial has to pass through the endpoints of each interval: $s_i(x_i) = y_i$ and $s_i(x_i + 1) = y_{i+1}$
- A boundary condition is imposed to determine that the polynomial coefficients have a unique solution; otherwise, the system of coefficients would have free variables, therefore not a unique solution.

(b) Data points $(-1, 1/2), (0, 1), (1, 2)$ and boundary condition $s'(-1) = 0$ to construct the piecewise quadratic interpolant.

1st piece: $s_0(x)$ on $[-1, 0]$

Formula: $s_0(x) = a_0 + b_0(x + 1) + c_0(x + 1)^2$

using: $(-1, 1/2)$: $s_0(-1) = a_0 = 1/2$

Using the boundary condition $s'_0(-1) = 0$: $s'_0(x) = b_0 + 2c_0(x + 1) \implies s'_0(-1) = b_0 = 0$

Using the next point $(0, 1)$: $s_0(0) = \frac{1}{2} + 0(1) + c_0(1)^2 = 1 \implies c_0 = 1/2$

Therefore the first piece of the interpolant is: $s_0(x) = \frac{1}{2} + \frac{1}{2}(x + 1)^2$

2nd piece $s_1(x)$ on $[0, 1]$

Formula: $s_1(x) = a_1 + b_1(x - 0) + c_1(x - 0)^2$

using: $(0, 1)$: $s_1(0) = a_1 = 1$.

To find b_1 we will use the fact that the derivative at 0 has to be continuous so $s'_0(x) = s'_1(x)$ So

$x = 0$: $s'_0(x) = x + 1 \implies s'_0(0) = 1$

$s'_1(x) = b_1 + 2c_1x \implies s'_1(0) = b_1$

Therefore, $b_1 = 1$

Using the point $(1, 2)$: $s_1(1) = 1 + 1(1) + c_1(1)^2 = 2 \implies c_1 = 0$

So the second piece of the interpolant is: $s_1(x) = 1 + x$

the final piecewise function is :

$$s(x) = \begin{cases} \frac{1}{2} + \frac{1}{2}(x + 1)^2, & x \in [-1, 0] \\ 1 + x, & x \in [0, 1] \end{cases}$$

(c) Since we established that the first derivative is continuous at $x = 0$ from the initial conditions, we can take either polynomial piece to evaluate. Picking $s_1(x) = 1 + x \implies s'_1(x) = 1$. So the estimated derivative

at $x = 0$ is equal to 1.

Evaluating $f'(x) = 2^x \ln 2$ at $x = 0 \implies f'(0) = 2^0 \ln 2 = \ln 2$

Since the piecewise polynomial function is an interpolant of this function, its derivative should be an approximation of $f'(0)$ of the original function. This is not a great example of an approximation since we calculated $s'_1(x) = 1$, but $f'(0) \approx 0.69$

(d) If we look at the way our data points are spaced out, we see that for $x+1$, the y value doubles and we know that the exponential function doubles for every $x+1$ value. Instead of forcing the slope at the points to be equal to 0, we can force it to be some specific number. Using the exponential function, we can say that the slope will be:

$$f'(0) = 2 \cdot f'(-1)$$

$$f'(1) = 2 \cdot f'(0)$$

$f'(1) = 4 \cdot f'(-1)$ I am not sure if this is the right approach since when I calculate these values with the polynomial, my approximation is equal to 0.66...

Exercise 3

(a) Neville's method for points: $(0, 2)$, $(1, -1)$, and $(2, 4)$ to evaluate at $\bar{x} = 1.3$

The formula for this method :

$$P_{i,j}(\bar{x}) = \frac{(\bar{x} - x_{i-j})P_{i-j+1,i}(\bar{x}) - (\bar{x} - x_i)P_{i-j,i-1}(\bar{x})}{x_i - x_{i-j}}$$

Degree 0 :

$$P_0(1.3) = 2$$

$$P_1(1.3) = -1$$

$$P_2(1.3) = 4$$

Degree 1 :

$$P_{0,1}(1.3) = \frac{(1.3-1)(2)-(1.3-0)(-1)}{0-1} = \frac{0.6+1.3}{-1} = -1.9$$

$$P_{1,2}(1.3) = \frac{(1.3-2)(-1)-(1.3-1)(4)}{1-2} = \frac{0.7-1.2}{-1} = 0.5$$

Degree 2 :

$$P_{0,1,2}(1.3) = \frac{(1.3-2)(-1.9)-(1.3-0)(0.5)}{0-2} = \frac{1.33-0.65}{-2} = -0.34$$

So the approximate value at $\bar{x} = 1.3$ using Neville's method is equal to -0.34

(b) Find the same interpolating polynomial using Newton's form :

Newton's Polynomial Form:

$$P(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \dots + a_n(x - x_0) \dots (x - x_{n-1})$$

The coefficients:

$$a_0 = f[x_0]$$

$$a_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = f[x_0, x_1]$$

$$a_2 = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = f[x_0, x_1, x_2]$$

First coefficient: $a_0 = f[x_0] = 2$

Second coefficient: $a_1 = f[x_0, x_1] = -3$

Third coefficient: $a_2 = f[x_0, x_1, x_2] = 4$

The resulting polynomial will be :

$$P_2(x) = 2 + (-3)(x - 0) + 4(x - 0)(x - 1)$$

$$P_2(x) = 2 - 3x + 4x(x - 1)$$

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$
0	2	-3	4
1	-1		
		5	
2	4		

Table 1: Newton's Divided Difference Table for Exercise 3(b)

Plugging in $\bar{x} = 1.3$:

$$P(1.3) = 2 - 3(1.3) + 4(1.3)(1.3 - 1)$$

$$P(1.3) = 2 - 3.9 + 4(1.3)(0.3)$$

$$P(1.3) = -1.9 + 1.56 = -0.34$$

So the approximate value at $\bar{x} = 1.3$ using Newton's method is equal to -0.34

Exercise 4

The output of the calculation of the L_∞ and L_2 norm of the interpolation error gives us the output:

uniform: Linf = 0.076727, L2 = 0.063187

chebyshev: Linf = 0.046107, L2 = 0.044128

legendre: Linf = 0.092126, L2 = 0.039161

The L_∞ norm is the maximum error at every single point and it gives us the worst case scenario, while the L_2 norm is the root mean square error and it gives us the average error across the interval. Here, the Chebyshev points have the lowest norms, which is what we expected since it works with the min-max logic, so they are spaced out in a way that minimizes the maximum possible error. The Uniformly spaced points had the worst overall output since they are evenly spaced. This means that the end points will shoot off or under to overcompensate to reach the actual values we are trying to get to. So, somehow counter-intuitively the uniformly spaced points don't end up performing really well. The Legendre points are have the lowest L_2 norm and the highest L_∞ , which is the expected behavior. The L_2 norm is the lowest since these points make the integral(total area) of the squared error as small as possible and the L_∞ is high since they are fitted as close to the function on an average scale across the entire interval. (I am not sure why its higher than the uniform ones?)

I think that the usage of each of these spaced out points vary if we increase the degree of the polynomial.

Exercise 5

(a) (ii) Centered first derivative:

Stencil: $\{x_0 - h, x_0, x_0 + h\}$

System: $f'(x_0) \approx a \cdot f(x_0 - h) + b \cdot f(x_0) + c \cdot f(x_0 + h)$

Taylor Expansion:

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2 + \frac{f'''(x_0)}{6}h^3 + \dots$$

$$f(x_0 - h) = f(x_0) - f'(x_0)h + \frac{f''(x_0)}{2}h^2 - \frac{f'''(x_0)}{6}h^3 + \dots$$

$$f'(x_0) \approx (a + b + c)f(x_0) + (c - a)hf'(x_0) + (a + c)\frac{h^2}{2}f''(x_0) + \dots$$

Solve the system:

$$f(x_0): a + b + c = 0$$

$$f'(x_0): (c - a)h = 1 \implies c - a = \frac{1}{h}$$

$$f''(x_0): a + c = 0 \implies a = -c$$

$$\text{Substitute : } c - (-c) = \frac{1}{h} \implies 2c = \frac{1}{h} \implies c = \frac{1}{2h}$$

$$\text{Then } a = -\frac{1}{2h}$$

$$\text{Substituting back: } -\frac{1}{2h} + b + \frac{1}{2h} = 0 \implies b = 0$$

$$\text{Therefore: } f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h}$$

So the coefficients a, b and c are the ones we expect to see in the output of our code.

(iii) Centered second derivative:

$$\text{Stencil: } \{x_0 - h, x_0, x_0 + h\}$$

Taylor Expansion:

$$f''(x_0) \approx (a + b + c)f(x_0) + (c - a)hf'(x_0) + (a + c)\frac{h^2}{2}f''(x_0) + (c - a)\frac{h^3}{6}f'''(x_0) + \dots$$

Solve the system:

$$f(x_0): a + b + c = 0$$

$$f'(x_0): c - a = 0 \implies a = c$$

$$f''(x_0): (a + c)\frac{h^2}{2} = 1 \implies a + c = \frac{2}{h^2}$$

$$\text{Now we have } c + c = \frac{2}{h^2} \implies 2c = \frac{2}{h^2} \implies c = \frac{1}{h^2}$$

$$\text{Since } a = c, a = \frac{1}{h^2}$$

$$\text{and } \frac{1}{h^2} + b + \frac{1}{h^2} = 0 \implies b = -\frac{2}{h^2}$$

$$\text{Therefore: } f''(x_0) = \frac{f(x_0-h) - 2f(x_0) + f(x_0+h)}{h^2}$$

So the coefficients a, b and c are the ones we expect to see in the output of our code.

(i) Forward first derivative:

$$\text{Stencil: } \{x_0, x_0 + h\}$$

$$f'(x_0) \approx A \cdot f(x_0) + B \cdot f(x_0 + h)$$

$$\text{Taylor expansion: } f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2 + \dots$$

$$f'(x_0) \approx a(f(x_0)) + b(f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2 + \dots)$$

$$f'(x_0) \approx (a + b)f(x_0) + (b \cdot h)f'(x_0) + (b \cdot \frac{h^2}{2})f''(x_0) + \dots$$

Solve the system:

$$f(x_0): a + b = 0$$

$$f'(x_0): b \cdot h = 1$$

$$b = \frac{1}{h}$$

$$a + \frac{1}{h} = 0 \implies a = -\frac{1}{h}$$

$$\text{Therefore: } f'(x_0) \approx -\frac{1}{h}f(x_0) + \frac{1}{h}f(x_0 + h) = \frac{f(x_0+h) - f(x_0)}{h}$$

So the coefficients a and b are the ones we expect to see in the output of our code.

Exercise 6

(a) In order to prove that D is equal to that expression, we will apply Taylor's expansion on $f(x_0 - h)$ about x_0 . So :

$$f(x_0 - h) = f(x_0) - hf'(x_0) + \frac{h^2}{2!}f''(x_0) - \frac{h^3}{3!}f'''(x_0) + \frac{h^4}{4!}f^{(4)}(x_0) - \frac{h^5}{5!}f^{(5)}(x_0) + O(h^6)$$

rewriting the expression to get the difference $f(x_0) - f(x_0 - h)$:

$$f(x_0) - f(x_0 - h) = hf'(x_0) - \frac{h^2}{2}f''(x_0) + \frac{h^3}{6}f'''(x_0) - \frac{h^4}{24}f^{(4)}(x_0) + \frac{h^5}{120}f^{(5)}(x_0) + O(h^6)$$

so dividing by h to get the term D_h :

$$D_h = \frac{hf'(x_0) - \frac{h^2}{2}f''(x_0) + \frac{h^3}{6}f'''(x_0) - \frac{h^4}{24}f^{(4)}(x_0) + \frac{h^5}{120}f^{(5)}(x_0) + O(h^6)}{h} \implies$$

$$D_h = f'(x_0) - \frac{h}{2}f''(x_0) + \frac{h^2}{6}f'''(x_0) - \frac{h^3}{24}f^{(4)}(x_0) + \frac{h^4}{120}f^{(5)}(x_0) + o(h^4)$$

Since we want to use the small- o notation, we can say $O(h^5)$ is "contained" within $o(h^4)$ (since h^5 goes to zero faster than h^4 as h goes to 0) and that is why we replaced in the Taylor's expression. Finally,

$$f'(x_0) = D = D_h + \left(\frac{f''(x_0)}{2}\right)h + \left(-\frac{f'''(x_0)}{6}\right)h^2 + \left(\frac{f^{(4)}(x_0)}{24}\right)h^3 + \left(-\frac{f^{(5)}(x_0)}{120}\right)h^4 + o(h^4)$$

where k_i 's are the constants in front of h.

$$(b) f(x) = \ln(x^2 + 1)$$

Calculate derivatives:

1st derivative:

$$f'(x) = \frac{2x}{x^2+1}$$

2nd derivative:

$$f''(x) = \frac{2(1-x^2)}{(x^2+1)^2}$$

3rd derivative:

$$f'''(x) = \frac{4x(x^2-3)}{(x^2+1)^3}$$

4th derivative:

$$f^{(4)}(x) = \frac{-12(x^4-6x^2+1)}{(x^2+1)^4}$$

5th derivative:

$$f^{(5)}(x) = \frac{48x(x^4-10x^2+5)}{(x^2+1)^5}$$

Since we want to evaluate at $x_0 = 1$, we will plug it into each derivative:

$$f''(1) = \frac{2(1-1)}{(1+1)^2} = 0$$

$$f'''(1) = \frac{4(1)(1-3)}{(1+1)^3} = \frac{-8}{8} = -1$$

$$f^{(4)}(1) = \frac{-12(1-6+1)}{(1+1)^4} = \frac{-12(-4)}{16} = \frac{48}{16} = 3$$

$$f^{(5)}(1) = \frac{48(1)(1-10+5)}{(1+1)^5} = \frac{48(-4)}{32} = \frac{-192}{32} = -6$$

Using the formula from (a) we get:

$$D = D_h + 0h + \frac{1}{6}h^2 + \frac{1}{8}h^3 + \frac{1}{20}h^4 + o(h^4)$$

(c) Starting from $h=1$, we want to approximate the first derivative of $f(x) = \ln(x^2 + 1)$ at $x_0 = 1$, by extrapolation on D_h . $D_h = \frac{f(1)-f(1-h)}{h}$, where $f(1) = \ln(2) \approx 0.69314718$

For extrapolation, we must evaluate at $\frac{h}{2}$ at each step so starting at $h=1$:

For $h = 1.0$: $N_1(1) = \frac{\ln(2) - \ln(1)}{1} \approx 0.69314718$
For $h = 0.5$: $N_1(0.5) = \frac{\ln(2) - \ln(1.25)}{0.5} \approx 0.94000726$
For $h = 0.25$: $N_1(0.25) = \frac{\ln(2) - \ln(1.5625)}{0.25} \approx 0.98744032$
For $h = 0.125$: $N_1(0.125) = \frac{\ln(2) - \ln(1.765625)}{0.125} \approx 0.99738008$

From our notes we know the general formula $N_{j+1}(h) = \frac{2^p N_j(h/2) - N_j(h)}{2^p - 1}$.
Extrapolating N_2 :
 $N_2(1)$: $\frac{2^1 N_1(0.5) - N_1(1)}{2^1 - 1} = 2(0.94000726) - 0.69314718 \approx 1.18686734$
 $N_2(0.5)$: $2(0.98744032) - 0.94000726 \approx 1.03487338$
 $N_2(0.25)$: $2(0.99738008) - 0.98744032 \approx 1.00731984$

Extrapolating N_3 :
 $N_3(1)$: $\frac{2^2 N_2(0.5) - N_2(1)}{2^2 - 1} = \frac{4(1.03487338) - 1.18686734}{3} \approx 0.98420873$
 $N_3(0.5)$: $\frac{4(1.00731984) - 1.03487338}{3} \approx 0.99813533$

Extrapolating N_4 :
 $N_4(1)$: $\frac{2^3 N_3(0.5) - N_3(1)}{2^3 - 1} = \frac{8(0.99813533) - 0.98420873}{7} \approx 1.00012484$

h	$N_1(h) (O(h))$	$N_2(h) (O(h^2))$	$N_3(h) (O(h^3))$	$N_4(h) (O(h^4))$
1.0	0.69314718			
0.5	0.94000726	1.18686734		
0.25	0.98744032	1.03487338	0.98420873	
0.125	0.99738008	1.00731984	0.99813533	1.00012484

Table 2: Extrapolation Table for $f'(1)$ where $f(x) = \ln(x^2 + 1)$

(besides our notes, this is the reference I used, especially on how to build the table Extrapolation).

(d) Since the true derivative is $D = f'(1) = 1$, the absolute error for the final extrapolated value $N_4(1)$ will be:

$$|Error| = |1 - 1.00012484| = 1.2484 \times 10^{-4}$$

Since each extrapolation deletes the leading error term from the Taylor series expansion, the fourth column N_4 achieves a theoretical convergence rate of $O(h^4)$. To find the step size needed to achieve the same error without extrapolation, we take first order backwards approximation that we calculated on (b) but we'll only keep the first term for simplicity:

$$\begin{aligned} \text{Error} &= \frac{1}{6} h^2 \implies \\ \frac{1}{6} h^2 &= 1.2484 \times 10^{-4} \\ h &= \sqrt{7.4904 \times 10^{-4}} \approx 0.02737 \end{aligned}$$

So we would need to use a step size of roughly $h \approx 0.027$, which is much smaller than the $h = 1$ step we used in extrapolation that only required evaluating the function 4 times. Therefore, the extrapolation method is much quicker.

Sources

<https://math.berkeley.edu/~mgu/MA128ASpring2017/MA128ALectureWeek6.pdf>